

CPS 150 Review Session #1

Key Concepts in Java

Important Ideas

IPO

General problem solving approach to computer programming

- **Input** - What is given by the user
- **Process** - How we manipulate the input to get a desired result
- **Output** - Returning the result of our process to the user

Java Essentials

Class Definition

```
public class MyClass { }
```

Main Method Header

```
public static void main(String[] args) { }
```

Java Essentials

Inline Comments

```
// this is a comment
```

Block Comments

```
/*  
    this  
    is  
    a  
    comment  
*/
```

Print Statements

How to output information to the user

Examples

```
System.out.print("Hello World");
```

```
System.out.println("Hello World");
```

Variables

How to store information

Examples in Java

```
int myVar1 = 5;  
double myVar2 = 4.162;
```

Arithmetic

Math class

```
Math.pow(2.0, 3.0);  
Math.sqrt(4.0);  
Math.round(5.123);  
Math.random();
```

Arithmetic

Division

```
// int vs double division
int divInt = 5 / 2;
double divDouble = 5.0 / 2.0;

// divide by zero
int divZero1 = 5 / 0;
double divZero2 = 5.0 / 0.0;

// long division (division vs modulo)
int myNumQuotient = 5 / 2;
int myNumRemainder = 5 % 2;
```

Arithmetic

Division

```
// int vs double division
int divInt = 5 / 2; // 5 / 2 = 2
double divDouble = 5.0 / 2.0; // 5.0 / 2.0 = 2.5

// divide by zero
int divZero1 = 5 / 0; // Runtime Error
double divZero2 = 5.0 / 0.0; // Infinity

// long division (division vs modulo)
int myNumQuotient = 5 / 2; // 2
int myNumRemainder = 5 % 2; // 1
```

Scanner

```
Scanner myScanner = new Scanner(System.in);
```

Putting It All Together

```
import java.util.Scanner;

public class Demo {
    public static void main(String args[]) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int num1 = input.nextInt();
        System.out.print("Enter second number: ");
        int num2 = input.nextInt();

        int sum = num1 + num2;
        System.out.println("The sum of the two numbers is " + sum);
    }
}
```

What's Wrong With This Code?

```
Public Class myClass {  
    public static void main(string args) {  
        system.out.println>Hello  
        system.out.println>Welcome to CPS 150)  
    }
```

What's Wrong With This Code?

```
public class myClass {  
    public static void main(String[] args) {  
        System.out.println("Hello");  
        System.out.println("Welcome to CPS 150");  
    }  
}
```

Write Your Own Programs

Write a program that gets a user input temperature in Celsius and converts it to Fahrenheit

```
Enter a temperature in degrees Celsius: 25  
The temperature in Fahrenheit is 77.0F
```

Write a program that gets a user input number of seconds and converts it to minutes and seconds

```
Enter a number of seconds: 125  
125 seconds is 2 minutes and 5 seconds
```